

Reproduction of pytorch/torch titan: Llama 3 debug-model pretraining on one GPU

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Independent reproduction · Artifact evaluation · Natural language processing

Abstract

Ran the README quick start at the pinned commit on a single GPU: run_train.sh with NGPU=1 and the shipped llama3_debugmodel recipe (small Llama 3 architecture, the c4_test sample data bundled in the repository, AdamW lr 8e-4 with linear decay), extended from 10 to 2,000 steps through a two-line recipe wrapper (kv_recipes.py) because recipe settings are no longer command-line options. PyTorch nightly 2.16.0.dev20261007 for CUDA 13.0, as the README requires a nightly build. One NVIDIA RTX 5060 Ti.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Llama 3 debug model, 2000 st...	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: Llama 3 debug model, 2000 steps

Run	Metric	Final value	Step
1	loss	1.68376	2000
1	grad_norm	0.4213	
1	tflops	27.28	
1	mfu	8.74	
1	mean_tokens_per_second	204465	

Run	Inline claim	Value	Phase
1	first_loss	7.98317	
1	final_loss_mean_last100	1.7264	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	191.6 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	5db71f0f719da80f2b94d1d424f79fba656ebc0242e358b79161adc7643a3294	2.4 MB
r1.kvbundle.zip	9f8a20e45d71db5db150c421d2d32eccf42183e902e66d000b0e76348ec457a7	4.1 MB

5 Links and licence

Paper: <https://arxiv.org/abs/2410.06511>

Reproduction of: <https://github.com/pytorch/torchtitan/tree/f4ce10595537576d62513dacc382e6684dadf6f2>

Record: CC-BY 4.0. Code: BSD-3-Clause, as declared by the authors. Tags: llm-pretraining, torchtitan, llama3, distributed-training.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Llama 3 debug model, 2000 steps), kv:2610.00029v1/r1. Data hash 902ef28d56b69e5748e88bcc531d53db9a87875582631ae2eb9a8a81aae3c091.
<https://kveritas.org/records/2610.00029v1/r1.pdf>